

10. (3.75 points)

Given the following output from a generalized linear model fitted to a triangle of loss development data as of December 31, 2018:

Accident Year	Standardized Pearson Residuals as of (months)		
	12	24	36
2016	-0.54	1.05	3.56
2017	1.43	2.75	
2018	0.00		

Accident Year	Standard Deviation
2016	2.067
2017	0.933
Total	1.570

Accident Year	Fitted Cumulative Losses (\$000) as of (months)		
	12	24	36
2016	1,250	2,000	2,150
2017	1,500	2,400	
2018	1,400		

Accident Year	Sample Residual Index (row, column) as of (months)		
	12	24	36
2016	(2, 1)	(1, 1)	(2, 1)
2017	(1, 3)	(2, 2)	
2018	(1, 2)		

- Assume the residuals in accident years 2017 and 2018 are homoscedastic.

a. (2 points)

Construct the triangle of sampled cumulative losses based on the above sampling algorithm and adjusting the residuals as necessary.

Given the following additional information:

Cumulative Distribution Function	Expected Losses
20%	2,500
60%	3,000
100%	3,500

- The tail factor as of 36 months is 1.15.
- 0% loss trend.
- The first stochastically generated random seed for the expected loss is 0.458 and should be applied to the fitted loss triangle.
- The second stochastically generated random seed for the expected loss is 0.812 and should be applied to the sampled loss triangle.

b. (1.75 points)

Calculate the range of stochastic Bornhuetter-Ferguson reserve indications.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 10

TOTAL POINT VALUE: 3.75

LEARNING OBJECTIVE(S): A7, A8, A10

SAMPLE ANSWERS

Part a: 2 points

Sample 1:

AY	h_i
16	$1.57 / 2.067 = 0.7596$
17 - 18	$1.57 / 0.933 = 1.6827$

Adj Residuals = $r \times h_i$			
AY	12	24	36
16	-0.410	0.798	2.704
17	2.406	4.627	
18	0.000		

Sampled Incr. $m_{w,d} + r/h_i (m_{w,d})^{0.5}$			
AY	12	24	36
16	1362	735	189
17	1562	982	
18	1418		

Example Calc.	
AY16,12M	$1250 + (2.406 / 0.7596) \times (1250)^{0.5} = 1362$
AY17,12M	$1500 + (2.704 / 1.6827) \times (1500)^{0.5} = 1562$

Sampled Cumulative			
AY	12	24	36
16	1362	2097	2286
17	1562	2545	
18	1418		

Sample 2:

Fitted Incr.			
AY	12	24	36
16	1250	750	150
17	1500	900	
18	1400		

Residuals in 2016 & 2017/2018 are heteroscedastic

$$h_{16} = 1.57 / 2.067 = 0.76$$

$$h_{17/18} = 1.57 / 0.933 = 1.683$$

Sampled Residual		
12	24	36
$2.41 / 0.76 = 3.17$	-0.54	3.17
$2.71 / 1.683 = 1.61$	2.75	
$0.798 / 1.683 = 0.474$		

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Incremental		
12	24	36
$1250 + 3.17 \times (1250)^{0.5} = 1362$	$750 + -0.54 \times (750)^{0.5} = 735$	$150 + 3.17 \times (150)^{0.5} = 189$
$1500 + 1.61 \times (1500)^{0.5} = 1562$	$900 + 2.75 \times (900)^{0.5} = 983$	
$1400 + 0.474 \times (1400)^{0.5} = 1418$		

Cumulative		
12	24	36
1362	2097	2286
1562	2545	
1418		

Sample 3:

This calculation is the same as the calculation from Solution 1, but is done in \$000's instead of taking the values as shown.

Fitted Incre.			
AY	12	24	36
16	1250000	750000	150000
17	1500000	900000	
18	1400000		

Residuals in 2016 & 2017 are heteroscedastic

$$h_{16} = 1.57 / 2.067 = 0.76$$

$$h_{17} = 1.57 / 0.933 = 1.683$$

Sampled Residual		
12	24	36
$2.41 / 0.76 = 3.17$	-0.54	3.17
$2.71 / 1.683 = 1.61$	2.75	
$0.798 / 1.683 = 0.474$		

Incremental		
12	24	36
$1250000 + 3.17 \times (1250000)^{0.5} = 1,253,542$	$750000 + -0.54 \times (750000)^{0.5} = 749,532$	$150000 + 3.17 \times (150000)^{0.5} = 151,227$
$1500000 + 1.61 \times (1500000)^{0.5} = 1,501,968$	$900000 + 2.75 \times (900000)^{0.5} = 902,609$	
$1400000 + 0.474 \times (1400000)^{0.5} = 1,400,561$		

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Cumulative		
12	24	36
1,253,542	2,003,074	2,154,301
1,501,968	2,404,577	
1,400,561		

Part b: 1.75 points

Sample 1:

First Sim

$$E\{L\} = 3000$$

$$LDF_{12-24} = 1.600$$

$$LDF_{24-36} = 1.075$$

$$CDF = 1.6 \times 1.075 \times 1.15 = 1.978$$

AY	E(L)	$(1 - p) = 1 - 1 / CDF$	CDF	R
16	3k	0.130	1.15	390
17	3k	0.191	1.236	573
18	3k	0.494	1.978	1482
			Total	2445

Second Sim

AY	E(L)	$(1 - p) = 1 - 1 / CDF$	CDF	R
16	3500	0.130	1.15	455
17	3500	0.203	1.254	710.5
18	3500	0.497	1.990	1739.5
			Total	2905

The range of stochastic BF reserve indications is 2,445,000 to 2,905,000

Sample 2:

First Sim

$$E(L) = 2500 + 500 \times [(0.458 - 0.200) / (0.6 - 0.2)] = 2822.5$$

$$LDF_{12-24} = 1.600$$

$$LDF_{24-36} = 1.075$$

$$CDF = 1.6 \times 1.075 \times 1.15 = 1.978$$

AY	E(L)	$(1 - p) = 1 - 1 / CDF$	CDF	R
16	2822.5	0.130	1.15	368
17	2822.5	0.191	1.236	539
18	2822.5	0.494	1.978	1396
			Total	2303

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Second Sim

$$E(L) = 3000 + 500 \times [(0.812 - 0.600) / (0.6 - 0.2)] = 3265$$

AY	E(L)	$(1 - p) = 1 - 1 / \text{CDF}$	CDF	R
16	3265	0.130	1.15	426
17	3265	0.203	1.254	661
18	3265	0.497	1.990	1624
			Total	2711

The range of stochastic BF reserve indications is 2,303 to 2,711

EXAMINER'S REPORT

Candidates were expected to adjust the development triangle with heteroscedastic data and use the fitted triangle and the result from part a) to calculate the range of BF method reserve indications.

Candidates had more difficulty with the calculation of the BF method reserve indications in part b) than the adjustment of the heteroscedastic data in part a).

Part a

Candidates were expected to note that the accident year standard deviations were significantly different for 2016 and 2017, and calculate the heteroscedastic adjustment factors to put the residuals on the same variance level so random sampling is appropriate. These factors needed to be correctly applied to the residual triangle to calculate the incremental sampled triangle that was then used to calculate the cumulative sampled triangle.

Common mistakes included:

- Incorrectly adjusting the residual triangle
- Skipping the calculation of the incremental fitted triangle and going straight for the cumulative sampled triangle.
- Not readjusting the variance when calculating the sample incremental triangle (dividing by the appropriate heteroscedasticity adjustment factor for the sampled accident year)
- Adjusting 2016 to the total variance level and stating that 2017 and/or 2018 did not require a heteroscedasticity adjustment.
- Not calculating the cumulative triangle and leaving the calculation at the incremental sample triangle.

Part b

Candidates were expected to use the provided seed information to look up the expected losses, calculate the LDFs / CDFs for both the fitted and sampled triangle from part a), then calculate the BF reserve for each accident year and sum to find the total range of reserves.

Common mistakes included:

- Using the incorrect expected loss on the fitted / sampled triangles
- Using LDFs from only one of the fitted / sampled triangles to calculate the range for both simulations
- Incorrectly applying the BF methodology.

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

- Missing the tail factor when calculating the cumulative loss development factor
- Calculating the range for each individual accident year, but not calculating the total overall range of reserves.
- Misusing the information provided to estimate the range of reserves.